MILITARY SPECIFICATION

POTASSIUM BIFLUORIDE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of potassium bifluoride (KHF_2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60	-	Tape: Packaging, Waterproof

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-117 - Bags, Sleeves and Tubing-Interior Packaging
MIL-W-10842 - Wax, Microcrystalline

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical characteristics. Potassium bifluoride shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as KHF_2)	99.0	---	4.2.4.1
Chloride (Cl)	---	0.02	4.2.4.2
Iron (Fe)	---	0.05	4.2.4.3
Potassium fluosilicate (K_2SiF_6)	---	0.5	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the potassium bifluoride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For potassium bifluoride test (see 6.5). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 50 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for potassium bifluoride test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Interior lining of microcrystalline wax missing or not as specified
(f) Fiberboard pads, separators, and sleeves missing or not as specified
(g) Unitization not as specified
(h) Marking incorrect, missing, or illegible

4.2.3.2 For potassium bifluoride test (see 6.5). Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

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4.2.4 Potassium bifluoride tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as KHF_2).

(a) Zirconium-alizarin mixture. Dissolve 1.0 g of sodium alizarin sulfonate in 100 milliliters (ml) of 95-percent ethanol. Filter off the residue and add an additional 150 ml of 95-percent ethanol to the filtrate. Dissolve 1.00 g of zirconium nitrate, $Zr(NO_3)_4 \cdot 5H_2O$, in 250 ml of water. Mix the two solutions as needed using 2 parts of the ethanol solution to 3 parts of the aqueous solution.

(b) Standard sodium fluoride solution. Dissolve 0.4200 g of sodium fluoride (which has been purified by recrystallizing, washing, and igniting strongly) in water, volumetrically, to make 100 ml of solution. This solution is 0.1N.

(c) Standardized 0.1N thorium nitrate solution. Standardize 0.1N thorium nitrate solution by titrating 9.00 ml of standard sodium fluoride solution prepared as specified in (b) which has been transferred by use of a microburet into a 100-ml tall-form beaker by using the procedure specified in (d) below, beginning with "Add sufficient water to make a total volume of approximately 20 ml...."

(d) Procedure. Weigh to the nearest 0.1 milligram (mg) approximately 0.90 g of the specimen into a 50-ml distillation flask. Add a few glass beads, 5 ml of 60-percent perchloric acid, and sufficient water to cause the solution to boil at no more than 110°C. Place the flask on an asbestos mat with an opening large enough so that about one-third of the flask will be exposed to the flame. Close the flask with a two-hole rubber stopper through which passes a thermometer and a capillary tube, both of which extend down into the liquid. Connect a dropping funnel to the capillary tube and fill with water. Attach a water condenser to the distillation apparatus and collect the distillate in an open container. Allow the temperature of the distillation to reach a maximum of 135°C and then maintain this temperature by the addition of water from the dropping funnel. Distill over 75.0 ml of distillate and dilute to 250 ml with water in a volumetric flask. Transfer 10.0 ml of this solution into a 100 ml tall-form beaker. Add sufficient water to make a total volume of approximately 20 ml and 3 drops of zirconium-alizarin mixture prepared as specified in (a). If necessary, add just enough 1 to 50 hydrochloric acid to decolorize the solution. Add an equal volume of neutral ethanol and titrate over a white surface with thorium nitrate solution prepared and standardized as specified in (d) to a faint reappearance of color. (The reaction is slow near the end point.) Calculate the percent by weight potassium bifluoride as follows:

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$$\text{Percent by weight potassium bifluoride} = \frac{97.63\ AB}{W}$$

where: A = Milliliters of thorium nitrate solution used,
B = Normality of thorium nitrate solution, and
W = Weight of specimen in grams.

4.2.4.2 Chloride. Weigh 0.100 g of the specimen and dissolve in 20 ml of water. Add 0.20 g of boric acid, 1.0 ml of 70-percent nitric acid, and 1.0 ml of 2-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced in a control containing 0.02 mg of chloride ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

4.2.4.3 Iron.

(a) Butanolic potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in 10 ml of water. Warm the solution to 25° to 30°C and add n-butanol to make 100 ml of solution. Shake vigorously until clear.

(b) Procedure. Weigh 2.000 g of the specimen and evaporate with 5 ml of 37-percent hydrochloric acid to dryness in a platinum dish on a steam bath. Repeat the evaporation with another 5 ml of 37-percent hydrochloric acid. Warm the residue with 2 drops of 37-percent hydrochloric acid and 10 ml of water. Dilute to 20 ml with water, and filter. To 10 ml of the filtrate, add 1 ml of 37-percent hydrochloric acid, 30 mg of ammonium persulfate, and 15 ml of butanolic potassium thiocyanate solution prepared as specified in (a). Shake vigorously for 30 seconds and allow to separate. Any red color produced in the clear butanol layer shall be no darker than that produced when 0.5 mg of iron ion is tested in the same manner as the specimen.

4.2.4.4 Potassium fluosilicate (K_2SiF_6). Weigh 2.000 g of the specimen and dissolve in 40 ml of water in a platinum crucible. Add 10 ml of saturated potassium nitrate solution, cool to 0°C, and add 3 drops of phenolphthalein. Titrate with 1N sodium hydroxide solution until a pink color persists for 15 seconds. Heat the solution to 80° ± 2°C and titrate with 0.1N sodium hydroxide solution to a permanent pink color. No more than 1.9 ml of the 0.10N sodium hydroxide solution shall be necessary to produce the permanent pink color.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side
(d) On other side

Examine the container after each period for any evidence of leakage.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Potassium bifluoride shall be unit packed level A in a 1- or 25-pound (lb) quantity as specified (see 6.2).

(a) One-pound quantity. A quantity of 1 lb (+0.2 or -0 ounce) of potassium bifluoride shall be unit packed in a wide mouth glass bottle conforming to type I, class 1 of PPP-C-2020. The glass shall conform to type b, light penetration grade 2 of PPP-C-186. The interior surface of the bottle and neck shall be completely lined with a film of microcrystalline wax conforming to type I of MIL-W-10842. Alternatively, the potassium bifluoride shall be unit packed in a type III, class 3 polyethylene wide mouth bottle conforming to PPP-C-2020. Each bottle shall be closed with a polyethylene screw-cap, and secured by a plastic shrinkable band or tape. There shall be no evidence of leakage of contents when the bottle is tested as specified in 4.2.5.

(b) Twenty-five pound quantity. A quantity of 25 (+0.5 or -0) lb of potassium bifluoride shall be unit packed in a bag within a removable head polyethylene drum conforming to Department of Transportation (DOT) Specification 35. The bag shall conform to type I, class B, style 2 of MIL-B-117 and shall be formed from minimum nominal 0.005 inch thick polyethylene. The bag shall be closed by tying, heat sealing, or knotting. There shall be no evidence of leakage of contents when the drum is tested as specified in 4.2.5.

5.1.1.2 Intermediate packing. Twelve bottles of potassium bifluoride, unit packed as specified in 5.1.1.1, shall be intermediately packed in a fiberboard box conforming to style RSC, type CF, grade W5c of PPP-B-636. Arrangement of the bottles shall be in three rows of four bottles each, and shall be separated from one another by means of half-slotted fiberboard separators designed to form a full-bottle-height close-fitting cell for each bottle. The box shall be furnished with a sleeve and top and bottom pads. Additional pads shall be inserted wherever needed to prevent motion of contents. Separators, sleeves, and pads shall be formed from fiberboard conforming to grade W5c of PPP-F-320. Closure of the fiberboard box shall be in accordance with the requirements in the appendix to PPP-B-636, using tape conforming to PPP-T-60.

5.1.2 Level C.

5.1.2.1 Unit packing. Potassium bifluoride shall be unit packed level C in a 1- or 25-lb quantity as specified (see 6.2).

(a) One-pound quantity. A quantity of 1 lb (+0.2 or -0 ounce) of potassium bifluoride shall be unit packed level C in a container and in a manner which shall neither affect the specified purity of or be affected by the contents, for a minimum period of 6 months.

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(b) Twenty-five-pound quantity. A quantity of 25 (+0.5 or -0) lb of potassium bifluoride shall be unit packed level C in a manner and in a container to assure maintenance of the specified purity of contents for a minimum period of 6 months, and to assure carrier acceptance and safe delivery from supply source to first destination. Packing shall be in compliance with applicable regulatory requirements and in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules applicable to any other selected mode of transportaion.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-pound quantity. Four intermediate packs, prepared as specified in 5.1.1.2, shall be packed in either a nailed wood, a cleated plywood, or a wirebound wood box conforming, respectively, to the following specifications: class 2, style 4 or 4-1/2 of PPP-B-621; type overseas, style A, B, I, or J of PPP-B-601; and class 3, style optional of PPP-B-585. Fiberboard filler pads of material conforming to grade W5c of PPP-F-320 shall completely separate the contents from the shipping container. Additional pads shall be inserted where needed to prevent motion of contents within the box. Each shipping container shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.1.2 Twenty-five-pound quantity. Potassium bifluoride, unit packed as specified in 5.1.1.3, shall require no further protection for shipment other than unitization.

5.2.2 Level B.

5.2.2.1 One-pound quantity. Potassium bifluoride shall be packed level B in the same manner as for level A, except that the shipping container shall be a fiberboard box instead of a wood box. The fiberboard box shall conform to grade V3c of PPP-B-636.

5.2.2.2 Twenty-five-pound quantity. The twenty-five-pound quantity of potassium bifluoride shall be packed level B in the same manner as for level A protection.

5.2.3 Level C.

5.2.3.1 One-pound quantity. Twenty-four bottles of potassium bifluoride shall be packed in a fiberboard box in accordance with Uniform Freight Classification Rules and National Motor Freight Classification Rules, to assure carrier acceptance, to assure safe delivery from supply source to first destination, and to assure protection of contents for a minimum period of six months.

5.2.3.2 Twenty-five-pound quantity. The 25-lb quantity, unit packed level C as specified in 5.1.2, shall require no further protection for shipment.

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5.3 Unitization. Packs of potassium bifluoride shall be palletized in accordance with MIL-STD-147 using requirements for load type I for boxes, and those for load type III for pails. The pallet shall conform to NN-P-71, type IV.

5.4 Marking. Containers and pallet loads shall be marked in accordance with MIL-STD-129. In addition, each container and pallet load shall be marked to show the lot number and date of manufacture of the potassium bifluoride, and the following precautionary information:

HAZARDS

Highly Toxic by Ingestion
Corrosive, Eye and Skin

DANGER! MAY BE FATAL IF SWALLOWED

Wash thoroughly after handling

POISON
Call a physician

DANGER! CAUSES SEVERE BURNS
Do not get in eyes, on skin, or on clothing
Use chemical resistant gloves and aprons
Avoid breathing dust and vapor
Keep container closed
Keep away from heat or acid
Use with adequate ventilation
Clean up spillage

FIRST AID:

If swallowed, and the victim is conscious, give large quantities of water or other consumable liquids that are readily available. Do not attempt to induce vomiting. Seek medical care immediately.

In case of contact, immediately flush eyes or skin with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Call a physician. Wash clothing before re-use. (Discard contaminated shoes.)

6. NOTES

6.1 Intended use. Potassium bifluoride is intended for use in glass etching, in the manufacture of antiseptics and pickling solutions, and as a laboratory reagent.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation, unit packing, and packing required (see 5.1, 5.1.1, and 5.2)
(c) Unit quantity required (see 5.1.1.1 and 5.1.2.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially dangerous to personnel. Potassium biofluoride is highly toxic by ingestion, corrosive to eyes and skin. All applicable safety rules, regulations, and procedures must be followed in the sampling and testing of this material.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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